Decomposing VaR

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Executive Summary

We outline a generic framework under which to model Value-at-Risk ("VaR") over a one-year horizon, as required by Solvency UK and Solvency II in Europe. We decompose risks into different distinct categories, starting from first principles, and in particular that the model must coincide with the production of the balance sheet.

1 Introduction

Under Solvency II, firms are required to hold capital (the Solvency Capital Requirement, or SCR) which ensures that their basic own funds are positive with a 99.5% probability over a one-year horizon. This brief note outlines a general framework for modelling risk in a one-year VaR framework and defines the different sources of risk with mathematical clarity under such a general setting. The concepts that we introduce are not necessarily new. Our aim however is to remove ambiguity and to recommend a common framework under which which, ignoring any operational constraints, all specialisms inside the Life Actuarial team can operate.

The rest of the paper continues as follows. In Section 2, we give a general description of the environment; in Section 3, we define the risks excluding model risk; in Section 4, we define model risk and in Section 5, we provide examples of how this framework applies to various product types.

2 A general description of the environment

Every year, the firm must estimate its technical provisions given the information available to it at that point in time. We define technical provisions as the value of funds required to transfer those liabilities ('transfer value').

2.1 Information set

We denote the information set as $I_t \subset I$ as being the set of all available information, or the 'global' information set, at time t, where I is the set of all possible information sets (the 'information universe'). It is then clear that $I_t \subset I_{t+j} \forall j \geq 0$ (i.e. any information known in a given year is also known in all subsequent years). We also denote the set $i_t \subset I_t$ as the set of information *known to the firm*, or the 'local' information set, at time t.

2.2 The firm's liability portfolio

We denote the portfolio set $P_t \subset P$ as the set describing the portfolio characteristics of the firm, or the portfolio-specific information, where, as above, P is the set of all possible portfolios (the 'portfolio universe').

2.3 Information choice function

Given this, we denote the set function $g_t : i_t \to i_t$ as the 'information choice function' used within the firm which maps the local information set onto a subset of that. This subset is the information which is used by the firm as an input into its valuation model (which can include expert judgement - see Section 2.4). In other words, this function represents the process used within the firm to select the information used to help formulate a value for the firm's liabilities. Keeping notation consistent, we define g as the set of all possible information choice functions, so that $g_t \subset g$.

2.4 Valuation

We can now define the set function $f_t : P \times I \to \mathbb{R}$ as the function at time t which gives the transfer value of the liabilities for portfolio P_t given the information set I_t.

We distinguish between 'true' VaR and 'estimated' VaR. We make this distinction since we can never know for certain the true probability law underlying a random process and therefore a firm's VaR can only be an estimate of the true VaR for that risk and as such is a random variable itself. Under this framework, we can then make explicit statements about model risk by comparing the true VaR to a firm's estimated VaR.

2.5 Specifying 'true' VaR

In order to specify the true one-year VaR, we first need to define the 'true' transfer value of a firm's technical provisions in one year's time, given the global and the portfolio-specific information at that moment in time; consistent with the notation above, this is given as $f_{t+1}\left(P_{t+1}, I_{t+1}\right)$. We also define the function $c : P \times P \to \mathbb{R}$ as the one-year cashflow of the firm. For notational brevity, we define $c_t \in \mathbb{R}$ as $c_t = c\left(P_t, P_{t+1}\right)$ and the random variable $\tilde{c}_t = c\left(P_t, \tilde{P}_{t+1}\right)$

The problem we are then faced with is to find a function $g : P \times I \times [0,1] \to \mathbb{R}$ which, at any given time t, produces the risk capital for a one-year VaR for a portfolio $P_t \subset P$ and information set $I_t \subset I$ at the given confidence level $k \in [0,1]$ (e.g. $k = 0.995$ as in the ICA/SCR). This implies that the probability K of remaining solvent over the run-off of the portfolio (i.e. the probability of paying all liabilities in full over the lifetime of the portfolio) is given as $K = (1-k)^T$, where T is the term of the run-off[1].

2.6 Solution to the problem

We now define the function $g : P \times I \times [0,1] \to \mathbb{R}$ as the function which, at time t, produces the firm's risk capital for portfolio $P_t \subset P$ and information set $I_t \subset I$ at the given confidence level $k \in [0,1]$.

[1]Note that these conditions combined are stronger than simply the condition that the probability of non-ruin over the portfolio run-off is K, which can be achieved by setting the probability of non-ruin over some years to be lower than k and over other years to be higher than k

Definition 1 ($A(y)$)**.** *The multivalued function $A : \mathbb{R} \to \mathbb{R} \times I \times F \times F$ produces a set of values such that:*

$$[\tilde{c}_t - \mathrm{E}[c_t]] + \left[f\left(\tilde{I}_{t+1}\right) - \mathrm{E}\left[f\left(I_{t+1} \mid I_t\right)\right]\right] \leq k \tag{1}$$

$$\Pr\left\{\left([\tilde{c}_t - \mathrm{E}[c_t]] + \left[f\left(\tilde{I}_{t+1}\right) - \mathrm{E}\left[f\left(I_{t+1} \mid I_t\right)\right]\right]\right) \cdot (1 + r_t)^{-1} \geq g(P_t, I_t, k)\right\} = k \tag{2}$$

where $r_t \in \mathbb{R}^+$ is the one-year spot risk-free rate of interest at time t. The SCR at time t is therefore given as $g(P_t, I_t, 0.005)$

$$\left(\hat{c}^*, \hat{I}^*_{t+1}, \hat{f}^*_{t+1}, \hat{F}^*_{t+1}\right) \in \underset{c, I_{t+1}, f_{t+1}, F_{t+1}}{\arg\min} \left([\tilde{c}_t - \mathrm{E}[c_t]] + \left[f\left(\tilde{I}_{t+1}\right) - \mathrm{E}\left[f\left(I_{t+1} \mid I_t\right)\right]\right]\right) \cdot (1 + r_t)^{-1} \geq g(P_t, I_t, k) \tag{3}$$

3 Decomposition of risks excluding model risk

3.1 Payout risk

Payout risk is the risk that the actual policyholders are different to those currently expected.

Definition 2 (Payout risk)**.** *The risk that:* $\tilde{c}_t - \mathrm{E}[c_t] > 0$

3.2 Recalibration risk

Recalibration risk is the risk that the firm's best estimate provisions in the next time period are different to those currently expected.

Definition 3 (Recalibration risk)**.** *The risk that:* $\tilde{f}_{t+1}\left(\tilde{I}_{t+1}\right) - \mathrm{E}\left[f\left(I_{t+1} \mid I_t\right)\right] > 0$

3.3 Provisionining risk

Provisioning risk is the risk that the firm's best estimate provisions in the next time period are different to those expected by the market and hence different to the transfer value of the firm's liabilities.

Definition 4 (Provisioning risk)**.** *The risk that:* $\tilde{F}\left(\tilde{I}_{t+1}\right) - f\left(\tilde{I}_{t+1}\right) > 0$

4 Model risk

Model risk is the risk that the firm's estimate of the distribution of the change in surplus

Definition 5 (Model risk)**.** *The risk that* $g(P_t, I_t, k) - \hat{g}(P_t, I_t, k) > 0$

Model risk can itself be decomposed into the following components:

- the risk of misestimating the one-year cashflow distribution: $\tilde{c}_t - \hat{c}^*_t > 0$
- the risk of misestimating the distribution of reserves in the next time period: $\tilde{f}_{t+1}\left(\tilde{I}_{t+1}\right) - \hat{f}^*_{t+1}\left(\hat{I}^*_{t+1}\right) > 0$, which itself consists of:

– the risk of misestimating the distribution of functions used at time $t+1$: $\tilde{f}\left(\tilde{I}_{t+1}\right) - f^*\left(\tilde{I}_{t+1}\right) > 0$

– the risk of misetimating the distribution of information at time $t+1$: $\hat{f}^*_{t+1}\left(\tilde{I}_{t+1}\right) - \hat{f}^*_{t+1}\left(\hat{I}^*_{t+1}\right) > 0$

"Uncertainty must be taken in a sense radically distinct from the familiar notion of Risk, from which it has never been properly separated.... The essential fact is that 'risk' means in some cases a quantity susceptible of measurement, while at other times it is something distinctly not of this character; and there are far-reaching and crucial differences in the bearings of the phenomena depending on which of the two is really present and operating.... It will appear that a measurable uncertainty, or 'risk' proper, as we shall use the term, is so far different from an unmeasurable one that it is not in effect an uncertainty at all." **- Knight, F.H. (1921)** [2].

"We were seeing things that were 25-standard deviation moves, several days in a row." **- David Viniar (2007)** [3].

"[T]here are known knowns; there are things we know we know. We also know there are known unknowns; that is to say we know there are some things we do not know. But there are also unknown unknowns – there are things we do not know we don't know." **- Donald Rumsfeld (2002)** [4].

[2]Risk, Uncertainty, and Profit. Boston, MA: Hart, Schaffner & Marx; Houghton Mifflin Company (source: `http://en.wikipedia.org/wiki/Knightian_uncertainty)`

[3]Goldman Sachs Chief Financial Officer, 13 August 2007 FT article (source: `http://www.ft.com/cms/s/0/d2121cb6-49cb-11dc-9ffe-0000779fd2ac.html#axzz1rr6JoJuJ)`

[4]United States Secretary of Defence, February 2002 press briefing (source: `http://en.wikipedia.org/wiki/There_are_known_knowns)`